

Hadamard Differentiability Without the Pain

A working recipe, a toolkit, and Leibniz rules for integrals

Companion to *Statistics for Insurance*, Chapter 1 (Bladt)

1 The idea in one sentence

Hadamard differentiability is ordinary differentiation along a line $\varepsilon \mapsto \phi(\theta + \varepsilon h)$, made *uniform* over the index t and *robust* to the direction wobbling ($h_n \rightarrow h$). So you **compute** the derivative with calculus, and you **justify** it with known building blocks and the chain rule.

Definition 1.12, restated

$\phi : \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}$ is Hadamard differentiable at θ (tangentially to $\mathbb{D}_0$) if there is a continuous linear ϕ'_θ with

$$\left\| \frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} - \phi'_\theta(h) \right\|_{\mathbb{E}} \rightarrow 0 \quad \text{for all } t_n \rightarrow 0, h_n \rightarrow h \in \mathbb{D}_0, \theta + t_n h_n \in \mathbb{D}_\phi.$$

Three things to remember: the limit is in **sup norm** (uniform in t), the derivative must be **linear in h** , and the direction h_n is allowed to **move**.

2 The three-step recipe

Step 1: Find the candidate derivative (pure calculus)

Fix t , treat ε as a real number, and compute

$$\phi'_\theta(h)(t) = \frac{d}{d\varepsilon} \phi(\theta + \varepsilon h)(t) \Big|_{\varepsilon=0}.$$

Use every rule from first-year calculus: chain rule, product rule, quotient rule, and the Leibniz rules of Section 4 for integrals. If ϕ is Hadamard differentiable, this *is* the derivative. **Sanity check:** the answer must be linear in h (no h^2 , no $|h|$, no $\sup h$).

Step 2: Decompose into building blocks

Write $\phi = \phi_k \circ \dots \circ \phi_1$ where each ϕ_j is in the toolkit (Section 3), then cite the **chain rule (Theorem 1.16)**:

$$(\psi \circ \phi)'_\theta = \psi'_{\phi(\theta)} \circ \phi'_\theta.$$

Push the direction through from the inside out: each block receives the *point* from the previous block and the *direction* from the previous derivative.

Step 3: Check the side conditions

This is where the actual Hadamard content lives. For every block, verify the condition in the last column of the toolkit table: bounded range, denominators away from zero, tangential directions continuous, bounded variation, etc. Only if a block is genuinely new do you go back to Definition 1.12 and do a t_n, h_n argument.

3 The toolkit

Throughout, F, G are points, h, k directions, all in ℓ^∞ with the sup norm.

Block	Derivative in direction h (and k)	Calculus analogue	Source	Side condition
Linear map L (integral $\int_0^\cdot$, evaluation, sums)	$L(h)$	$(ax)' = a$	Prop. 1.2(1–2)	L bounded: $\ Lh\ \leq C\ h\ _\infty$
Pointwise $F \mapsto \varphi(F(\cdot))$	$\varphi'(F(\cdot))h(\cdot)$	$\varphi(x)' = \varphi'(x)$	Lemma 1.10(2)	range of F in a bounded interval (a, b) where φ' is bounded and uniformly continuous
Product $F \cdot G$	$hG + Fk$	product rule	direct	none (uses $\sup_n \ h_n\ _\infty < \infty$)
Ratio F/G	$\frac{h}{G} - \frac{Fk}{G^2}$	quotient rule	Prop. 1.2(3)	$\inf_t G(t) > 0$
Inverse $F \mapsto F^{-1}$	$-(h/f) \circ F^{-1}$	$(f^{-1})' = \frac{1}{f' \circ f^{-1}}$	Lemma 1.10(1)	$f > 0$ continuous; only tangentially to $C[a, b]$
Composition $(F_1, F_2) \mapsto F_2 \circ F_1$	$h_2 \circ F_1 + F_{2, F_1}'(h_1)$	chain rule, both parts moving	Lemma 1.10(3)	h_2 uniformly continuous; F_2 uniformly Fréchet
$\int_{(-\infty, \cdot]} \psi(F_1) dF_2$	$[h_2 \psi(F_1)]_{-\infty}^\cdot - \int h_2(s-) d\psi(F_1(s)) + \int \psi'(F_1)h_1 dF_2$	Leibniz + integration by parts	Lemma 1.11	$\psi \in C^2$; $F_1, F_2 \in BV_M$
Product integral $\prod(1 + dF)$	$\int_{(0, \cdot]} \phi(F)(0, u-) \times \phi(F)(u, \cdot] dh(u)$	$(e^x)' = e^x$ (Duhamel)	Lemma 1.12–1.13	$F \in BV_M$

Always finish with the functional δ -method (Theorem 1.17): if $r_n(X_n - \theta) \rightsquigarrow X$ with X tight and in $\mathbb{D}_0$, then $r_n(\phi(X_n) - \phi(\theta)) \rightsquigarrow \phi'_\theta(X)$.

4 Leibniz rules for integrals

These are the tools for Step 1 whenever an integral appears.

4.1 Classical Leibniz rule

Leibniz integral rule

$$\frac{d}{d\varepsilon} \int_{a(\varepsilon)}^{b(\varepsilon)} g(x, \varepsilon) dx = g(b(\varepsilon), \varepsilon) b'(\varepsilon) - g(a(\varepsilon), \varepsilon) a'(\varepsilon) + \int_{a(\varepsilon)}^{b(\varepsilon)} \partial_\varepsilon g(x, \varepsilon) dx.$$

Sufficient conditions: a, b differentiable; g and $\partial_\varepsilon g$ continuous in (x, ε) on a compact region.
On unbounded domains: $|\partial_\varepsilon g(x, \varepsilon)| \leq D(x)$ with $\int D < \infty$ (dominated convergence).

Read it as: *(value at the moving top) × (speed of top) − (value at the moving bottom) × (speed of bottom) + (differentiate inside).*

4.2 Special cases you will use constantly

1. Fixed limits: differentiate under the integral.

$$\frac{d}{d\varepsilon} \int_a^b g(x, \varepsilon) dx = \int_a^b \partial_\varepsilon g(x, \varepsilon) dx.$$

2. Moving upper limit only (fundamental theorem of calculus).

$$\frac{d}{d\varepsilon} \int_a^{b(\varepsilon)} g(x) dx = g(b(\varepsilon)) b'(\varepsilon).$$

3. Moving lower limit only.

$$\frac{d}{d\varepsilon} \int_{a(\varepsilon)}^b g(x) dx = -g(a(\varepsilon)) a'(\varepsilon).$$

4. Integrand is a function of the path (the functional case). For $\psi \in C^1$,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_0^t \psi(A(s) + \varepsilon h(s)) ds = \int_0^t \psi'(A(s)) h(s) ds.$$

This is “pointwise block, then linear integral block” and is Hadamard via Lemma 1.10(2) + Prop. 1.2(2) + chain rule.

4.3 Stieltjes integrals: perturbing both the integrand and the integrator

When both pieces move, apply the product rule to “integrand” and “measure” separately:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int \psi(F_1 + \varepsilon h_1) d(F_2 + \varepsilon h_2) = \underbrace{\int \psi'(F_1) h_1 dF_2}_{\text{integrand moves}} + \underbrace{\int \psi(F_1) dh_2}_{\text{measure moves}}.$$

Integration by parts (right-continuous, bounded variation)

For càdlàg U, V of bounded variation on $(a, b]$,

$$\int_{(a,b]} U(s) dV(s) = U(b)V(b) - U(a)V(a) - \int_{(a,b]} V(s-) dU(s).$$

Why this matters: in the δ -method the direction is a Gaussian limit such as $h_2 = \mathbb{B} \circ F$, which is *not* of bounded variation, so $\int \psi(F_1) dh_2$ makes no sense literally. Integration by parts moves the “d” onto the nice function $\psi(F_1)$:

$$\int_{(-\infty, x]} \psi(F_1) dh_2 = [\psi(F_1)h_2]_{-\infty}^x - \int_{(-\infty, x]} h_2(s-) d\psi(F_1(s)).$$

That is exactly the formula in Lemma 1.11.

4.4 A limit that depends on the function

When the integration limit is itself a functional of F (typically a quantile $q = F^{-1}(\alpha)$), combine the Leibniz boundary term with the inverse-map derivative

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} (F + \varepsilon h)^{-1}(\alpha) = -\frac{h(q)}{f(q)}.$$

See Example 3 below.

5 Worked examples

Example 1: $\phi(A)(t) = \exp(\int_0^t A(s)^2 ds)$ on $\ell^\infty([0, \tau])$

Step 1 (calculus).

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \exp\left(\int_0^t (A + \varepsilon h)^2\right) = e^{\int_0^t A^2} \cdot \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_0^t (A + \varepsilon h)^2 = e^{\int_0^t A^2} \int_0^t 2A(s)h(s) ds.$$

Linear in h . ✓

Step 2 (decompose). $\phi = \Psi_{\text{exp}} \circ \Phi \circ \Psi_{\text{sq}}$ with

- $\Psi_{\text{sq}}(A) = A^2$ pointwise, Lemma 1.10(2), derivative $h \mapsto 2Ah$;
- $\Phi(G) = \int_0^\cdot G$ linear, Prop. 1.2(2), derivative $h \mapsto \int_0^\cdot h$;
- $\Psi_{\text{exp}}(I) = e^I$ pointwise, Lemma 1.10(2), derivative $h \mapsto e^I h$.

Push the direction through: $h \rightarrow 2Ah \rightarrow \int_0^\cdot 2Ah \rightarrow e^{\int_0^\cdot A^2} \int_0^\cdot 2Ah$.

Step 3 (side conditions). If $\|A\|_\infty \leq M$: squaring on $(-M-1, M+1)$ and exp on $(-1, \tau M^2 + 1)$ both have bounded, uniformly continuous derivatives; perturbations $A + t_n h_n$ eventually stay inside since $\|t_n h_n\|_\infty \rightarrow 0$. Φ is bounded with $\|\Phi h\| \leq \tau \|h\|_\infty$. Done.

Example 2: cumulative hazard $\Lambda = -\log(1 - F)$

Step 1. $\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} [-\log(1 - F - \varepsilon h)] = \frac{h}{1 - F}.$

Step 2. Single pointwise block, $\varphi(x) = -\log(1 - x)$.

Step 3. On $[0, \tau]$ with $F(\tau) < 1$, the range of F lies in $[0, F(\tau)] \subset (-1, 1)$, where $\varphi'(x) = 1/(1 - x)$ is bounded and uniformly continuous.

Conclusion. With $\sqrt{n}(F^{(n)} - F) \rightsquigarrow \mathbb{B} \circ F$,

$$\sqrt{n}(\Lambda^{(n)} - \Lambda) \rightsquigarrow \frac{\mathbb{B} \circ F}{1 - F} \text{ in } \ell^\infty([0, \tau]).$$

Example 3: tail integral with a quantile limit

$\psi(F) = \int_q^\infty x \, dF(x)$ with $q = F^{-1}(\alpha)$, F with density $f > 0$ near q , direction h continuous with $x h(x) \rightarrow 0$ as $x \rightarrow \infty$.

Step 1. Two effects, by the Leibniz rule.

- *Moving lower limit:* $-q f(q) \cdot (-h(q)/f(q)) = q h(q)$.
- *Moving measure:* $\int_q^\infty x \, dh(x) = [xh(x)]_q^\infty - \int_q^\infty h(x) \, dx = -q h(q) - \int_q^\infty h(x) \, dx$ (integration by parts).

Adding: $\psi'_F(h) = -\int_q^\infty h(x) \, dx$. The boundary terms cancel.

Step 2–3. Inverse block (Lemma 1.10(1), tangentially to continuous h), integral block (Lemma 1.11), chain rule. Truncate at a finite upper point, as in Examples 1.13–1.14, to stay inside the lemmas' assumptions.

Example 4: a new block, proved from the definition (product map)

$\phi(F, G) = FG$. Take $t_n \rightarrow 0$, $h_n \rightarrow h$, $k_n \rightarrow k$, and let $K = \sup_n (\|h_n\|_\infty \vee \|k_n\|_\infty) < \infty$. Then

$$\frac{(F + t_n h_n)(G + t_n k_n) - FG}{t_n} - (hG + Fk) = (h_n - h)G + F(k_n - k) + t_n h_n k_n,$$

with sup norm at most $\|G\|_\infty \|h_n - h\|_\infty + \|F\|_\infty \|k_n - k\|_\infty + t_n K^2 \rightarrow 0$. Three lines: this is what “direct” means in the table.

6 More examples

Unless stated otherwise, F is a distribution function, h (and k) are directions in ℓ^∞ , and in every δ -method conclusion $\sqrt{n}(F^{(n)} - F) \rightsquigarrow \mathbb{B} \circ F$ (Theorem 1.22). When F is supported on $[0, M]$, the relevant directions satisfy $h(0-) = h(M) = 0$, since they are differences of distribution functions (or limits thereof).

Example 5: the odds map $O(F) = \frac{F}{1-F}$ on $[0, \tau]$, $F(\tau) < 1$

Step 1. $\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \frac{F + \varepsilon h}{1 - F - \varepsilon h} = \frac{h(1 - F) + Fh}{(1 - F)^2} = \frac{h}{(1 - F)^2}$.

Step 2. One pointwise block, $\varphi(x) = x/(1 - x)$, $\varphi'(x) = (1 - x)^{-2}$ (Lemma 1.10(2)).

Step 3. The range of F is $[0, F(\tau)]$. On $(a, b) = (-1, \frac{1+F(\tau)}{2})$, φ' is bounded and uniformly continuous.

Conclusion. $\sqrt{n}(O(F^{(n)}) - O(F)) \rightsquigarrow \frac{\mathbb{B} \circ F}{(1 - F)^2}$, with pointwise variance $\frac{F(1 - F)}{(1 - F)^4} = \frac{F}{(1 - F)^3}$.

Example 6: the stop-loss premium process $\pi_F(d) = \mathbb{E}[(X - d)_+] = \int_d^M (1 - F(x)) \, dx$, $d \in [0, M]$

Step 1. $\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_d^M (1 - F - \varepsilon h) \, dx = -\int_d^M h(x) \, dx$.

Step 2. The map is *affine* in F : a constant plus the linear map $F \mapsto -\int_0^M F$. For affine maps the difference quotient equals the derivative exactly, so there is no limit to control.

Step 3. Boundedness, uniformly in d : $\sup_d \left| \int_d^M h \right| \leq M \|h\|_\infty$.

Conclusion. $\sqrt{n}(\pi_{F^{(n)}} - \pi_F) \rightsquigarrow -\int_0^M \mathbb{B}(F(x)) dx$ in $\ell^\infty([0, M])$, a centred Gaussian process with

$$\text{Cov}(\pi(d_1), \pi(d_2)) = \int_{d_1}^M \int_{d_2}^M [F(x \wedge y) - F(x)F(y)] dy dx.$$

Example 7: joint life, $\phi(F, G) = 1 - (1 - F)(1 - G)$

If $X \sim F$ and $Y \sim G$ are independent lifetimes, $\phi(F, G)$ is the distribution function of $\min(X, Y)$.

Step 1. $\frac{d}{d\varepsilon} \Big|_{\varepsilon=0} [1 - (1 - F - \varepsilon h)(1 - G - \varepsilon k)] = h(1 - G) + (1 - F)k$.

Step 2. Affine maps $F \mapsto 1 - F$, $G \mapsto 1 - G$, then the product map (Example 4), then $x \mapsto 1 - x$. Chain rule.

Step 3. None beyond Example 4.

Conclusion. With two independent samples of size n and independent bridges $\mathbb{B}_1, \mathbb{B}_2$ (joint convergence by Lemma 1.8):

$$\sqrt{n}(\phi(F^{(n)}, G^{(n)}) - \phi(F, G)) \rightsquigarrow (1 - G) \mathbb{B}_1 \circ F + (1 - F) \mathbb{B}_2 \circ G.$$

Example 8: the Nelson–Aalen map $\Lambda(F)(t) = \int_{(0,t]} \frac{dF(s)}{1 - F(s)}$, F continuous, $t \in [0, \tau]$

Step 1. Perturb both the integrand and the measure (Section 4.3):

$$\frac{d}{d\varepsilon} \Big|_{\varepsilon=0} \int_0^t \frac{d(F + \varepsilon h)}{1 - F - \varepsilon h} = \int_0^t \frac{dh}{1 - F} + \int_0^t \frac{h}{(1 - F)^2} dF.$$

Integrate the first term by parts ($h(0) = 0$): $\int_0^t \frac{dh}{1 - F} = \frac{h(t)}{1 - F(t)} - \int_0^t h \frac{dF}{(1 - F)^2}$. The two integrals cancel:

$$\Lambda'_F(h)(t) = \frac{h(t)}{1 - F(t)}.$$

Step 2. Diagonal map $F \mapsto (F, F)$ (linear), then Lemma 1.11 with $\psi(x) = 1/(1 - x)$, $F_1 = F_2 = F$. Lemma 1.11's formula reproduces exactly the three terms above.

Step 3. ψ must be C^2 on the range: fine on a neighbourhood of $[0, F(\tau)]$ since $F(\tau) < 1$. Bounded variation: F is monotone and bounded.

Sanity check. For continuous F , $\Lambda = -\log(1 - F)$, and Example 2 gave the same derivative $h/(1 - F)$. ✓

Example 9: the variance functional, F supported on $[0, M]$

$$\sigma^2(F) = \int_0^M x^2 dF - \left(\int_0^M x dF \right)^2 = L_2(F) - L_1(F)^2.$$

Step 1. Integration by parts with $h(0-) = h(M) = 0$:

$$\frac{d}{d\varepsilon} \Big|_{\varepsilon=0} L_1(F + \varepsilon h) = \int_0^M x dh = - \int_0^M h(x) dx, \quad \frac{d}{d\varepsilon} \Big|_{\varepsilon=0} L_2(F + \varepsilon h) = - \int_0^M 2x h(x) dx.$$

With $\mu = L_1(F)$, the outer map $(a, b) \mapsto b - a^2$ has gradient $(-2a, 1)$, so

$$(\sigma^2)'_F(h) = - \int_0^M 2x h \, dx + 2\mu \int_0^M h \, dx = -2 \int_0^M (x - \mu) h(x) \, dx.$$

Step 2. By parts, $L_1(F) = M - \int_0^M F \, dx$ and $L_2(F) = M^2 - \int_0^M 2xF(x) \, dx$: both affine, bounded (Prop. 1.2(1)). Then the smooth finite-dimensional map $(a, b) \mapsto b - a^2$, which is Hadamard (Fréchet) differentiable on $\mathbb{R}^2$. Chain rule.

Step 3. Continuity: $|(\sigma^2)'_F(h)| \leq 2M^2 \|h\|_\infty$.

Conclusion. $\sqrt{n}(\hat{\sigma}_n^2 - \sigma^2) \rightsquigarrow -2 \int_0^M (x - \mu) \mathbb{B}(F(x)) \, dx$, a normal limit. This recovers Proposition 1.1 through the functional route.

Example 10: the mean excess process $e_F(u) = \frac{\int_u^M (1 - F(x)) \, dx}{1 - F(u)}$, $u \in [0, u_0]$, $F(u_0) < 1$

Write $N(u) = \int_u^M (1 - F)$ and $D(u) = 1 - F(u)$, so $e_F = N/D$.

Step 1. $N'_F(h) = - \int_u^M h$ and $D'_F(h) = -h(u)$. Quotient rule:

$$e'_F(h)(u) = \frac{N'D - ND'}{D^2} = - \frac{\int_u^M h(x) \, dx}{1 - F(u)} + e_F(u) \frac{h(u)}{1 - F(u)}.$$

Step 2. N : affine block (Example 6). D : affine block. Ratio: Prop. 1.2(3)-type quotient rule (product map and pointwise $x \mapsto 1/x$). Chain rule.

Step 3. Denominator bounded away from zero: $\inf_{u \leq u_0} D(u) = 1 - F(u_0) > 0$. This is why the index set stops at u_0 .

Conclusion. A functional CLT for $\sqrt{n}(e_{F(n)} - e_F)$ in $\ell^\infty([0, u_0])$ with limit $e'_F(\mathbb{B} \circ F)$; this is Exercise 16 of the notes, and its value at a fixed u recovers Theorem 1.11.

Example 11: the P–P plot / ROC map $\phi(F, G)(p) = G(F^{-1}(p))$, $p \in [p_1, p_2]$

Step 1. With $q_p = F^{-1}(p)$ and $G' = g$:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} (G + \varepsilon k)((F + \varepsilon h)^{-1}(p)) = k(q_p) + g(q_p) \cdot \left(- \frac{h(q_p)}{f(q_p)} \right).$$

Step 2. $(F, G) \mapsto (F^{-1}, G)$ by Lemma 1.10(1)(i), then $(F_1, F_2) \mapsto F_2 \circ F_1$ by Lemma 1.10(3). Chain rule.

Step 3. F continuously differentiable with $f > 0$ around $[q_{p_1}, q_{p_2}]$; directions h continuous (tangential); G continuously differentiable with bounded, uniformly continuous g (gives uniform Fréchet differentiability); k uniformly continuous. All hold for Brownian-bridge limits.

Conclusion. With independent samples, $\sqrt{n}(\hat{\phi} - \phi)(p) \rightsquigarrow \mathbb{B}_2(G(q_p)) - \frac{g(q_p)}{f(q_p)} \mathbb{B}_1(p)$.

Example 12: the Laplace transform process $\mathcal{L}_F(s) = \int_{[0, M]} e^{-sx} \, dF(x)$, $s \in [0, T]$

Step 1. By parts with $F(0-) = 0$, $F(M) = 1$:

$$\mathcal{L}_F(s) = e^{-sM} + s \int_0^M e^{-sx} F(x) \, dx, \quad \mathcal{L}'_F(h)(s) = s \int_0^M e^{-sx} h(x) \, dx.$$

Step 2. Affine in F , so the difference quotient is *exactly* the derivative.

Step 3. Uniform boundedness in s : $\sup_{s \leq T} |s \int_0^M e^{-sx} h| \leq TM \|h\|_\infty$.

Lesson. Many “integral against dF ” functionals become affine in F after integration by parts. Then Hadamard differentiability is free; only the uniform bound needs checking.

Example 13: a ratio of Values at Risk, $R(F) = \frac{\text{VaR}_p}{\text{VaR}_r} = \frac{q_p}{q_r}$, $q_r > 0$

Step 1. Write $f_p = f(q_p)$, $f_r = f(q_r)$. Using $\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} q_p = -h(q_p)/f_p$ and the quotient rule,

$$R'_F(h) = -\frac{h(q_p)}{f_p q_r} + \frac{q_p h(q_r)}{f_r q_r^2}.$$

Step 2. Inverse map (Lemma 1.10(1)(i)) into $\ell^\infty([u_1, u_2]) \ni p, r$; evaluation at p and r (bounded linear map to $\mathbb{R}^2$); smooth map $(a, b) \mapsto a/b$ on $b > 0$. Chain rule.

Step 3. f continuous and positive near q_p, q_r ; directions continuous; $q_r > 0$.

Conclusion. $\sqrt{n}(\hat{R} - R) \rightarrow N(0, v)$ with, using $\text{Cov}(\mathbb{B}(p), \mathbb{B}(r)) = p \wedge r - pr$,

$$v = \frac{p(1-p)}{f_p^2 q_r^2} + \frac{q_p^2 r(1-r)}{f_r^2 q_r^4} - \frac{2q_p(p \wedge r - pr)}{f_p f_r q_r^3}.$$

Example 14: a round trip through the product integral, $S(F) = \prod_{(0, \cdot]} (1 - d\Lambda(F))$, F continuous

Here $\Lambda(F)$ is the Nelson–Aalen map of Example 8. Without censoring this is the Kaplan–Meier functional, and it equals $1 - F$.

Step 1. Let $\eta = \Lambda'_F(h) = h/(1 - F)$. Lemma 1.12 applied at the point $-\Lambda$ in direction $-\eta$:

$$S'_F(h)(t) = - \int_{(0, t]} \phi(-\Lambda)(0, u-) \phi(-\Lambda)(u, t] d\eta(u).$$

For continuous F , $\phi(-\Lambda)(0, u-) \phi(-\Lambda)(u, t] = \phi(-\Lambda)(0, t] = S(t) = 1 - F(t)$, so

$$S'_F(h)(t) = -(1 - F(t)) \eta(t) = -(1 - F(t)) \frac{h(t)}{1 - F(t)} = -h(t).$$

Step 2. $F \mapsto \Lambda(F)$ (Example 8), $\Lambda \mapsto -\Lambda$ (linear), product integral (Lemma 1.12). Chain rule.

Step 3. $F(\tau) < 1$ for Example 8, and Λ of bounded variation on $[0, \tau]$. For a rough direction like $\eta = \mathbb{B} \circ F/(1 - F)$, the $d\eta$ -integral is read via integration by parts, as in Lemma 1.13.

Sanity check. $S(F) = 1 - F$ is affine with derivative $-h$. The long route and the short route agree. ✓

7 Twenty ε -replacement drills

The replacement rule

Replace the **function** F (not the index t) by $F + \varepsilon h$ *everywhere it appears*: inside integrals, in denominators, inside inverses, in the integrator dF . With two input functions, use $F + \varepsilon h$ and $G + \varepsilon k$ simultaneously. Then

$$\phi'_F(h) = \left. \frac{d}{d\varepsilon} \phi(F + \varepsilon h) \right|_{\varepsilon=0}.$$

One fact to memorise for inverses: $\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} (F + \varepsilon h)^{-1}(p) = -\frac{h(q_p)}{f(q_p)}$ with $q_p = F^{-1}(p)$.

Each drill lists the map, the replaced expression, the derivative, and the results to cite (Step 2) with the side condition (Step 3). Unless stated otherwise $t \in [0, \tau]$, F is a distribution function (so its range is in $[0, 1]$), and for F supported on $[0, M]$ the directions satisfy $h(0-) = h(M) = 0$.

A. Pointwise maps (Lemma 1.10(2))

Drill 1: $\phi(F)(t) = F(t)^3$

Replace $(F(t) + \varepsilon h(t))^3 = F^3 + 3\varepsilon F^2 h + 3\varepsilon^2 F h^2 + \varepsilon^3 h^3$

Derivative $3F(t)^2 h(t)$ (coefficient of ε)

Justify Lemma 1.10(2) with $\varphi(x) = x^3$ on $(-1, 2) \supset [0, 1]$, where $\varphi'(x) = 3x^2$ is bounded and uniformly continuous.

Drill 2: $\phi(F)(t) = \frac{1}{1 + F(t)}$

Replace $\frac{1}{1 + F(t) + \varepsilon h(t)}$

Derivative $-\frac{h(t)}{(1 + F(t))^2}$

Justify Lemma 1.10(2) with $\varphi(x) = 1/(1 + x)$ on $(-\frac{1}{2}, 2)$; no singularity since $F \geq 0$, so no extra assumption.

Drill 3: $\phi(F)(t) = \log F(t)$, $t \in [a, b]$, $F(a) > 0$

Replace $\log(F(t) + \varepsilon h(t))$

Derivative $\frac{h(t)}{F(t)}$

Justify Lemma 1.10(2) with $\varphi = \log$ on $(F(a)/2, 2)$; $\varphi' = 1/x \leq 2/F(a)$ there. $F(a) > 0$ is essential: $1/x$ blows up at 0.

Drill 4: series system, $\phi(F)(t) = (1 - F(t))^k$, $k \in \mathbb{N}$

Replace $(1 - F(t) - \varepsilon h(t))^k$

Derivative $-k(1 - F(t))^{k-1} h(t)$

Justify Lemma 1.10(2) with $\varphi(x) = (1-x)^k$ on $(-1, 2)$, a polynomial, so φ' is bounded and uniformly continuous on bounded sets. Interpretation: survival of the minimum of k iid lifetimes.

Drill 5: logit, $\phi(F)(t) = \log \frac{F(t)}{1 - F(t)}$, $t \in [a, b]$, $0 < F(a) \leq F(b) < 1$

Replace $\log(F + \varepsilon h) - \log(1 - F - \varepsilon h)$

Derivative $\frac{h}{F} + \frac{h}{1 - F} = \frac{h(t)}{F(t)(1 - F(t))}$

Justify Lemma 1.10(2) with $\varphi(x) = \log \frac{x}{1-x}$ on $(\frac{F(a)}{2}, \frac{1+F(b)}{2})$, bounded away from both 0 and 1.

B. Evaluations and shifts (linear maps)

Drill 6: $\phi(F)(t) = F(t)F(2t)$, $F \in \ell^\infty([0, 2\tau])$

Replace $(F(t) + \varepsilon h(t))(F(2t) + \varepsilon h(2t))$

Derivative $h(t)F(2t) + F(t)h(2t)$

Justify Not pointwise: $F(2t)$ looks at a different point. The shift $S : F \mapsto F(2\cdot)$ from $\ell^\infty([0, 2\tau])$ to $\ell^\infty([0, \tau])$ is linear with $\|Sh\| \leq \|h\|$; then the product map (Example 4); chain rule.

Drill 7: excess distribution, $\phi(F)(t) = \frac{F(t) - F(u)}{1 - F(u)}$, $t \in [u, \tau]$, $F(u) < 1$

Replace $\frac{F(t) + \varepsilon h(t) - F(u) - \varepsilon h(u)}{1 - F(u) - \varepsilon h(u)}$

Derivative $\frac{h(t) - h(u)}{1 - F(u)} + \frac{(F(t) - F(u))h(u)}{(1 - F(u))^2}$

Justify Evaluation $F \mapsto F(u)$ is bounded linear into $\mathbb{R}$; numerator affine; denominator a scalar with $1 - F(u) > 0$; quotient rule (product map $+ x \mapsto 1/x$ on a bounded interval away from 0); chain rule.

C. Integrals of pointwise transforms

Drill 8: $\phi(F)(t) = \int_0^t e^{-F(s)} ds$

Replace $\int_0^t e^{-F(s)-\varepsilon h(s)} ds$

Derivative $-\int_0^t e^{-F(s)} h(s) ds$

Justify Lemma 1.10(2) with $\varphi(x) = e^{-x}$ on $(-1, 2)$, then Prop. 1.2(2) (cumulative integral, bounded by $\tau\|h\|$); chain rule.

Drill 9: survival from a hazard rate, $\phi(A)(t) = \exp\left(-\int_0^t A(s) ds\right)$, $A \in \ell^\infty([0, \tau])$

Replace $\exp\left(-\int_0^t (A + \varepsilon h)\right)$

Derivative $-\exp\left(-\int_0^t A\right) \int_0^t h(s) ds$

Justify Prop. 1.2(2), then Lemma 1.10(2) with exp on a bounded interval containing $[-\tau\|A\| - 1, \tau\|A\| + 1]$; chain rule. A may be any bounded function, not only a distribution function.

Drill 10: product integral, $\phi(A)(t) = \prod_{(0,t]} (1 + dA)$, A continuous, BV, $A(0) = 0$

Replace $\prod_{(0,t]} (1 + d(A + \varepsilon h)) = e^{A(t)+\varepsilon h(t)}$ (continuous case)

Derivative $e^{A(t)} h(t)$

Justify Lemma 1.12: $\int_{[0,t]} \phi(A)(0, u-) \phi(A)(u, t] dh(u) = e^{A(t)} \int_{[0,t]} dh = e^{A(t)} h(t)$ for continuous A and $h(0) = 0$. Bounded variation of A on $[0, \tau]$.

D. Integrals against dF : perturb the integrator

Drill 11: the mean, $\phi(F) = \int_{[0,M]} x dF(x)$

Replace $\int_{[0,M]} x d(F + \varepsilon h)(x) = \phi(F) + \varepsilon \int x dh$

Derivative $\int_{[0,M]} x dh = Mh(M) - \int_0^M h(x) dx = -\int_0^M h(x) dx$

Justify Integration by parts turns it into the affine map $F \mapsto MF(M) - \int_0^M F$: evaluation plus Prop. 1.2(1). Bound $2M\|h\|$. Affine $\Rightarrow$ Hadamard for free.

Drill 12: general moment, $\phi(F) = \int_{[0,M]} g(x) \, dF(x)$, $g \in C^1[0, M]$

Replace $\int g \, d(F + \varepsilon h)$

Derivative $\int g \, dh = g(M)h(M) - \int_0^M g'(x)h(x) \, dx = - \int_0^M g'(x)h(x) \, dx$

Justify Integration by parts (needs $g \in C^1$); affine in F ; bound $(|g(M)| + M\|g'\|_\infty)\|h\|$.
Example: $g(x) = e^{sx}$ gives the moment generating function.

Drill 13: two functions, $\phi(F, G)(t) = \int_{(0,t]} F(s) \, dG(s)$

Replace $\int_{(0,t]} (F + \varepsilon h) \, d(G + \varepsilon k)$

Derivative $\int_{(0,t]} h \, dG + \int_{(0,t]} F \, dk = \int_{(0,t]} h \, dG + F(t)k(t) - \int_{(0,t]} k(s-) \, dF(s)$

Justify Lemma 1.11 with $\psi(x) = x$ (which is C^2); $F, G \in BV_M$. The dk term is written by parts so that it makes sense for a rough limit $k = \mathbb{B} \circ G$.

Drill 14: self-integral, $\phi(F)(t) = \int_{(0,t]} F(s) \, dF(s)$, F continuous

Replace $\int_{(0,t]} (F + \varepsilon h) \, d(F + \varepsilon h)$

Derivative $\int h \, dF + F(t)h(t) - \int h(s-) \, dF(s) = F(t)h(t)$

Justify Diagonal map $F \mapsto (F, F)$ (linear) then Drill 13. **Sanity check:** for continuous F , $\phi(F) = F^2/2$ and $(F^2/2)' = Fh$. ✓

E. Quantiles (Lemma 1.10(1))

Drill 15: Value at Risk, $\phi(F) = F^{-1}(p) = q_p$

Replace $(F + \varepsilon h)^{-1}(p)$

Derivative $-\frac{h(q_p)}{f(q_p)}$

Justify Lemma 1.10(1)(i) into $\ell^\infty([u_1, u_2]) \ni p$, then evaluation at p (bounded linear). Needs f continuous and > 0 near q_p ; tangential to continuous h .

Drill 16: centred quantile process, $\phi(F)(p) = F^{-1}(p) - F^{-1}(\frac{1}{2})$, $p \in [p_1, p_2] \ni \frac{1}{2}$

Replace $(F + \varepsilon h)^{-1}(p) - (F + \varepsilon h)^{-1}(\frac{1}{2})$

Derivative $-\frac{h(q_p)}{f(q_p)} + \frac{h(q_{1/2})}{f(q_{1/2})}$

Justify Lemma 1.10(1)(i) on $[p_1, p_2]$; then the linear map $Q \mapsto Q - Q(\frac{1}{2})$, bounded by $2\|\cdot\|$; chain rule. Same side conditions as Drill 15.

Drill 17: $\phi(F) = \int_{\alpha}^{\beta} q_u^2 du$, $0 < \alpha < \beta < 1$

Replace $\int_{\alpha}^{\beta} [(F + \varepsilon h)^{-1}(u)]^2 du$

Derivative $-2 \int_{\alpha}^{\beta} q_u \frac{h(q_u)}{f(q_u)} du$

Justify Lemma 1.10(1)(i) on $[\alpha, \beta]$; Lemma 1.10(2) with x^2 on the bounded range $(q_{\alpha} - 1, q_{\beta} + 1)$; Prop. 1.2(1) with $w \equiv 1$; chain rule. Needs $f > 0$ continuous on $[q_{\alpha} - \epsilon, q_{\beta} + \epsilon]$.

F. Normalisations and distances

Drill 18: integrated tail distribution, $\phi(F)(t) = \frac{\int_0^t (1 - F)}{\int_0^{\tau} (1 - F)}$, $\mu_{\tau} = \int_0^{\tau} (1 - F) > 0$

Replace $\frac{\int_0^t (1 - F - \varepsilon h)}{\int_0^{\tau} (1 - F - \varepsilon h)}$

Derivative $-\frac{\int_0^t h}{\mu_{\tau}} + \frac{\int_0^t (1 - F) \cdot \int_0^{\tau} h}{\mu_{\tau}^2}$

Justify Numerator: affine (Prop. 1.2(2)); denominator: scalar affine (Prop. 1.2(1)), bounded away from 0; quotient rule with a scalar denominator; chain rule.

Drill 19: Cramér–von Mises distance, $\phi(F, G) = \int_0^M (F(x) - G(x))^2 dx$

Replace $\int_0^M (F - G + \varepsilon(h - k))^2 dx$

Derivative $2 \int_0^M (F - G)(h - k) dx$

Justify Linear map $(F, G) \mapsto F - G$; Lemma 1.10(2) with x^2 on $(-2, 2)$; Prop. 1.2(1); chain rule. **Warning:** if $F = G$ the derivative is 0, so the δ -method gives a degenerate limit; one needs the rate n and a second-order expansion instead.

G. A non-example

Drill 20: the supremum, $\phi(A) = \sup_{t \in [0,1]} A(t)$, at $A \equiv 0$

Replace	$\sup_t \varepsilon h(t)$. Take $h(t) = t - \frac{1}{2}$: for $\varepsilon > 0$ this is $\varepsilon \cdot \frac{1}{2}$; for $\varepsilon < 0$ it is $\varepsilon \cdot \inf h = \frac{ \varepsilon }{2}$. So $\phi(\varepsilon h) = \varepsilon /2$.
Derivative	Does not exist. $ \varepsilon /2$ is not differentiable at $\varepsilon = 0$: the one-sided derivatives are $+\frac{1}{2}$ and $-\frac{1}{2}$.
Justify	The ε -computation itself exposes the failure. Only a directional derivative $h \mapsto \sup_{t \in \arg \max A} h(t)$ exists, and it is not linear, so Theorem 1.17 does not apply. Lesson: if the ε -calculation produces $ \varepsilon $, max or sup, stop.

8 Pitfalls

Common mistakes

1. **Confusing point and direction.** The derivative is evaluated *at* A and applied *to* h . An answer like $h^2 e^{\int h^2}$ mixes them up and is not linear in h .
2. **Using Lemma 1.10(2) on a path-dependent map.** It only covers $\phi(F)(t) = \varphi(F(t))$. Anything with $\int_0^t$, $\sup_{s \leq t}$ or F^{-1} inside is *not* pointwise: split it into blocks first.
3. **Forgetting the bounded range.** x^2 , e^x , $\log$, $1/x$ are only fine on bounded intervals (and away from singularities). State the interval explicitly.
4. **Pointwise convergence is not enough.** The limit must be uniform in t ; that is the whole difference from the finite-dimensional δ -method.
5. **sup and max are traps.** $F \mapsto \sup_t F(t)$ is only directionally differentiable (the derivative $h \mapsto \sup_{t \in \arg \max} h(t)$ is not linear), so the standard functional δ -method does not apply.
6. **$\int \psi dh$ with rough h .** If the direction is a Brownian-bridge-type limit, rewrite the integral by parts before claiming anything.

9 Exam template

How to write it up

1. **Candidate.** “Formally, $\frac{d}{d\varepsilon} \phi(\theta + \varepsilon h)|_{\varepsilon=0} = \dots$, which is linear and continuous in h since $\|\phi'_\theta(h)\| \leq C\|h\|$.”
2. **Decomposition.** “Write $\phi = \phi_3 \circ \phi_2 \circ \phi_1$ where ϕ_1 is ... (Lemma/Prop. ...), ...”
3. **Conditions.** “Since $\|\theta\|_\infty \leq M$, the range lies in $(a, b) = \dots$, where φ' is bounded and uniformly continuous; ... is bounded away from zero, ...”
4. **Conclude.** “By Theorem 1.16, ϕ is Hadamard differentiable at θ (tangentially to $\mathbb{D}_0$) with derivative ...”
5. **Apply.** “By Theorem 1.17 and Theorem 1.22, $\sqrt{n}(\phi(F^{(n)}) - \phi(F)) \rightsquigarrow \phi'_F(\mathbb{B} \circ F)$.”